

DAY 4

Scenario Planning and Sensitivity Analysis

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

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Valuation models are only as credible as their assumptions. This chapter equips you to stress-test any financial model by building structured scenario plans and sensitivity analyses—two skills that separate exam-ready analysts from spreadsheet operators. You will learn to define base, bull, and bear cases for a DCF, isolate the inputs that drive the most value, construct one- and two-variable data tables, and present results in decision-ready dashboards. By the end, you can answer three questions every CFO asks: What is the value today? What if key assumptions change? What has to be true for this deal to work?

Scenario Planning Frameworks: Base, Bull, and Bear Cases

Scenario planning is not the same as toggling a single cell in your model. A true scenario bundles together every assumption that would logically move in the same direction under a given set of real-world conditions. When a SaaS company accelerates growth, it simultaneously demands more sales headcount, higher cloud infrastructure spend, expanded working capital to fund longer sales cycles, and often a lower discount rate as investors reward momentum with a higher multiple and lower perceived risk. Pulling only one lever while holding everything else constant produces an internally inconsistent picture—and an unreliable valuation.

The Three-Scenario Architecture

Base case anchors to management guidance and consensus estimates. It is your best single estimate of how the business will perform under ordinary conditions—neither heroic nor pessimistic. Think of it as the path the company is most likely to travel.

Bull case encodes a specific upside catalyst: a new product line gaining traction faster than expected, a competitor exiting the market, or a macro tailwind accelerating enterprise software budgets. The bull case is not simply "add five points to revenue growth." It must tell a coherent story where faster growth is funded by proportionally higher capex and working capital investment, and where the improved competitive position justifies a slightly lower WACC.

Bear case reflects a credible downside: customer churn rising, a pricing war compressing margins, or a credit tightening that raises the cost of capital. Crucially, lower revenue growth in the bear case also means lower capex needs—but margin compression may persist because fixed costs do not shrink as fast as revenue.

Common mistake: Analysts often build the base case first, then nudge numbers up or down to create the other two. This anchors the bull and bear cases to the base, producing scenarios that are too narrow to be useful. Build the bear case first, starting from a blank narrative of what could go wrong, before you have emotionally committed to any particular set of numbers.

Narrative Before Numbers

Before opening the spreadsheet, write two or three sentences explaining why each scenario is plausible. This discipline forces internal consistency. If your bull narrative says "enterprise deal sizes double," your model must reflect the longer implementation cycles, higher professional-services revenue, and increased accounts-receivable days that accompany large contracts.

Worked Example: SaaS Company DCF

Assume the company has \$100 million in current revenue, net debt of \$50 million, and 200 million diluted shares outstanding. We project five years of free cash flow, then apply a terminal value using a Gordon Growth Model at 3% perpetuity growth. For simplicity, unlevered free cash flow (UFCF) is approximated as $\text{EBITDA} \times (1 - \text{tax rate}) - \text{capex} - \text{change in working capital}$, with a 25% tax rate.

| Assumption | Bear | Base | Bull |

---|---|---|---

| Revenue CAGR (Yr 1–5) | 5% | 15% | 25% |

| EBITDA Margin | 12% | 20% | 25% |

| Capex (% of revenue) | 4% | 7% | 10% |

| 9Eworking Capital (% of rev 9B' Â RR Â ,R Â "R Â

| WACC | 12% | 10% | 8% |

| Scenario Probability | 25% | 50% | 25% |

Base case — Year 5 revenue: $\$100\text{M} \times (1.15)^5 = \201.1M . EBITDA = $\$201.1\text{M} \times 20\% = \40.2M . After-tax EBITDA = $\$40.2\text{M} \times 0.75 = \30.2M . Capex = $\$201.1\text{M} \times 7\% = \14.1M . 9Et2 \$, C ã Ò revenue increase $\times 8\% = \$8.1\text{M}$. Year-5 UFCF = $\$30.2\text{M} - \$14.1\text{M} - \$8.1\text{M} = \8.0M . Summing discounted UFCFs across all five years and adding a terminal value (Year-5 UFCF $\times 1.03 / (0.10 - 0.03)$) = $\$117.7\text{M}$, discounted at 10% for 5 years = $\$73.1\text{M}$) yields an enterprise value of approximately $\$310\text{M}$. Subtract net debt: equity value = $\$260\text{M} \div 200\text{M shares} = \1.30 per share.

Repeating the same mechanics:

- Bull case equity value per share: \$4.85
- Bear case equity value per share: \$0.42

Probability-Weighted Valuation

$$\text{Probability-Weighted Value} = P_{\text{base}} \times V_{\text{base}} + P_{\text{bull}} \times V_{\text{bull}} + P_{\text{bear}} \times V_{\text{bear}}$$

Applying the assigned probabilities:

$$\text{PWV} = (0.50 \times \$1.30) + (0.25 \times \$4.85) + (0.25 \times \$0.42)$$

$$\text{PWV} = \$0.65 + \$1.21 + \$0.11 = \$1.97 \text{ per share}$$

This \$1.97 figure is your central investment thesis price target—not the base case alone. The wide spread between \$0.42 and \$4.85 immediately signals that this is a high-variance investment where scenario selection, not just discount-rate precision, drives the conclusion.

Step-by-Step Procedure

- Write the narrative rationale for the bear case first; identify the specific risks driving each assumption lower.
- Write the bull case narrative; identify the catalysts and confirm that higher growth assumptions carry proportionally higher capex and working capital.
- Write the base case narrative aligned to management guidance and consensus.
- Populate all three sets of linked assumptions simultaneously—never copy-paste a single changed cell.
- Verify internal consistency: check that margin, capex, and working capital ratios move in the same direction as growth.
- Assign subjective probabilities that sum to 100%; document your rationale.

- Compute the probability-weighted value using the formula above.
- Stress-test the probabilities themselves: what happens to PWV if the bear probability rises to 40%?

The discipline of scenario planning is ultimately about intellectual honesty—acknowledging that the future is a distribution, not a point estimate, and pricing that uncertainty explicitly into your recommendation.

One-Variable Sensitivity Analysis and Tornado Charts

Scenario Planning Build Order



Build bear case first to avoid anchoring bull and base to optimism

One-variable sensitivity analysis answers a deceptively simple question: if only this one thing changes, how much does my valuation move? By isolating each input in turn, you build an honest map of where model risk actually lives — and that map is what a tornado chart displays.

The Core Procedure

Follow these steps for every input you want to test:

- Establish the base case. Lock every input at its central estimate and record the base-case output (e.g., enterprise value).
- Choose a range for the focal input. A standard convention is $\pm 10\text{--}20\%$ of the base-case value, though you may use analyst-defined bounds (e.g., WACC ± 300 bps around a 10% base).
- Vary the focal input across its range — typically in uniform steps — while holding all other inputs constant at their base-case values.
- Record the output at each step. You now have a series of output values corresponding to each input level.
- Calculate the output range as maximum output minus minimum output for that input. This single number captures the input's total leverage on the valuation.
- Repeat for every input you wish to test.
- Rank inputs by output range, largest to smallest.
- Plot a horizontal bar chart where each bar's length equals the output range, centered on the base-case value. The longest bar sits at the top; the shortest at the bottom. The resulting shape — wide at the top, narrow at the bottom — gives the tornado chart its name.

Worked Example: WACC vs. Revenue Growth Rate in a Manufacturing DCF

Suppose a manufacturing company has a base-case enterprise value (EV) of \$500 million, derived from a five-year DCF with a base WACC of 10% and a base terminal growth rate of 3%. You also want to test revenue growth rate sensitivity, with a base assumption of 5% annual growth.

Step 1 — Vary WACC from 7% to 13% in 1% steps.

| WACC | Enterprise Value |

|-----|-----|

| 7% | \$720 M |

| 8% | \$640 M |

| 9% | \$565 M |

| 10% | \$500 M (base) |

| 11% | \$445 M |

| 12% | \$398 M |

| 13% | \$358 M |

Output range for WACC = \$720 M " \$358 M = \$362 M

Step 2 — Vary revenue growth rate from 4% to 6% (±20% of the 5% base).

| Revenue Growth | Enterprise Value |

|-----|-----|

| 4% | \$455 M |

| 5% | \$500 M (base) |

| 6% | \$548 M |

Output range for revenue growth = \$548 M " \$455 M = \$93 M

Step 3 — Rank and plot. WACC's \$362 M range dwarfs revenue growth's \$93 M range, so WACC occupies the top bar of the tornado chart. Each bar extends left (low-input scenario) and right (high-input scenario) from the \$500 M center line.

Standardized Sensitivity Coefficients

Raw output ranges are useful but depend on how wide a range you chose for each input. To compare inputs on equal footing, compute the sensitivity coefficient:

$$\text{Sensitivity Coefficient} = (\Delta \text{Output} / \text{Output}_{\text{base}}) / (\Delta \text{Input} / \text{Input}_{\text{base}})$$

For WACC moving from 10% to 11% (a +10% change in the input):

- $\Delta \text{Output} / \text{Output}_{\text{base}} = (565 - 500) / 500 = 13.0\%$
- $\Delta \text{Input} / \text{Input}_{\text{base}} = (11 - 10) / 10 = 10.0\%$
- Sensitivity Coefficient = $13.0\% / 10.0\% = 1.30$

A coefficient of 1.30 means a 1% increase in WACC produces a 1.30% increase in enterprise value — slightly elastic. Running the same calculation for revenue growth typically yields a coefficient closer to 0.4 to 0.6, confirming WACC's dominance.

Why WACC and Terminal Growth Rate Dominate

In a DCF, the discount rate and terminal growth rate jointly determine the terminal value, which often represents 60–80% of total enterprise value. Small movements in either input therefore ripple through the largest component of the model. Revenue growth and margin assumptions matter, but their impact is diluted across only the explicit forecast period.

Common mistake: Analysts sometimes vary inputs by equal absolute amounts rather than proportional amounts, which artificially inflates the apparent sensitivity of inputs that happen to have large numerical values. Always normalize your range to a consistent percentage of the base-case value — or use the standardized coefficient — before ranking inputs and drawing conclusions about which drivers truly matter.

Communicating Risk with Tornado Charts

Tornado charts are among the most effective tools for translating quantitative model risk into a format non-technical stakeholders can absorb in seconds. A CFO or board member does not need to understand DCF mechanics to grasp that the WACC bar is four times longer than the revenue growth bar — and therefore that refinancing risk deserves far more attention than top-line forecast uncertainty. That instant visual hierarchy is the chart's primary value in practice.

Two-Variable Data Tables and Heat Maps

A two-variable data table is one of Excel's most powerful built-in tools for scenario planning because it lets you stress-test a valuation model across a full grid of input combinations in a single operation—no manual copy-pasting, no helper columns, no macros required. The result is a matrix of outputs that, once formatted as a heat map, communicates risk and upside at a glance to any audience, from a junior analyst to a board member.

How a Two-Variable Data Table Works

Excel's Data Table feature (Data !> What-If Analysis !> Data Table) requires exactly three ingredients: a row input cell, a column input cell, and a single formula output cell placed at the intersection of the row and column headers in your grid. Excel substitutes each row value into the row input cell and each column value into the column input cell simultaneously, recalculates the entire workbook, and records the output. The key constraint is that the output must be a single cell reference—if your valuation lives in a complex model, write a linking formula at the table's corner that points to the final output (e.g., enterprise value or equity value per share).

Common pairings for DCF work include WACC vs. terminal growth rate (the two inputs that dominate terminal value sensitivity) and revenue growth vs. EBITDA margin (which together drive free cash flow). Both pairings expose the inputs that analysts and clients argue about most, making them ideal for board presentations and fairness opinions.

The Core Formula

The terminal value drives the majority of DCF value in most models, so it is the natural focal point of a two-variable table:

$$\text{Terminal Value} = \text{FCF}_n \times (1 + g) / (\text{WACC} - g)$$

Small simultaneous moves in WACC and g create nonlinear, asymmetric swings in terminal value—exactly why a grid beats a one-at-a-time sensitivity table.

Building the 7×7 Retail DCF Table

Setup assumptions (base case):

- Projected Year 5 free cash flow (FCF ...): \$42 million
- Base WACC: 9.5%
- Base terminal growth rate (g): 2.5%
- Sum of PV of FCFs (Years 1–5): \$138 million
- Net debt: \$55 million; shares outstanding: 10 million

Step 1 — Calculate base-case terminal value.

$$TV = \$42M \times (1 + 0.025) / (0.095 - 0.025) = \$43.05M / 0.070 = \$615.0M$$

Step 2 — Discount terminal value to present (Year 0) at 9.5% for 5 years.

$$PV(TV) = \$615.0M / (1.095)^5 = \$615.0M / 1.5742 = \$390.7M$$

Step 3 — Compute enterprise value and equity value per share.

$$\text{Enterprise Value} = \$138M + \$390.7M = \$528.7M$$

$$\text{Equity Value} = \$528.7M - \$55M = \$473.7M$$

$$\text{Equity Value per Share} = \$473.7M / 10M = \$47.37$$

This \$47.37 is the formula output cell at the top-left corner of your data table.

Step 4 — Lay out the grid.

Place WACC values 8.0%, 8.5%, 9.0%, 9.5%, 10.0%, 10.5%, 11.0% across the row header (7 values).

Place terminal growth rates 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0% down the column header (7 values). The corner cell references your equity-value-per-share formula.

Step 5 — Run the data table.

Select the full 8×8 range (corner + headers + 49 output cells), open Data !' What-If Analysis !' Data Table, set the row input cell to your WACC assumption cell and the column input cell to your g assumption cell, and click OK. Excel populates all 49 cells instantly.

Resulting corner values (selected):

| WACC \ g | 1.0% | 2.5% | 4.0% |

| ---|---|---|---|

8.0%	\$44.91	\$58.62	\$83.47
9.5%	\$34.18	\$47.37	\$68.94
11.0%	\$26.05	\$37.22	\$55.61

Applying the Heat Map

Select the 49 output cells, go to Home !' Conditional Formatting !' Color Scales, and choose the Red–Yellow–Green three-color scale. Excel automatically assigns green to the highest values (low WACC, high growth), red to the lowest (high WACC, low growth), and yellow to the midrange. Circle or bold-border the base-case cell (\$47.37 at 9.5% / 2.5%) so readers immediately anchor to management's central estimate.

Identifying the Breakeven Contour

If the acquisition offer price is \$45.00 per share, scan the grid for cells nearest that value—this traces the breakeven contour, the combinations of WACC and g at which the DCF exactly justifies the offer. In the example above, the contour runs roughly from (8.5%, 1.5%) through (9.5%, 2.0%), giving the deal team a clear picture of how much growth and cost-of-capital cushion exists.

Keeping the Table Live

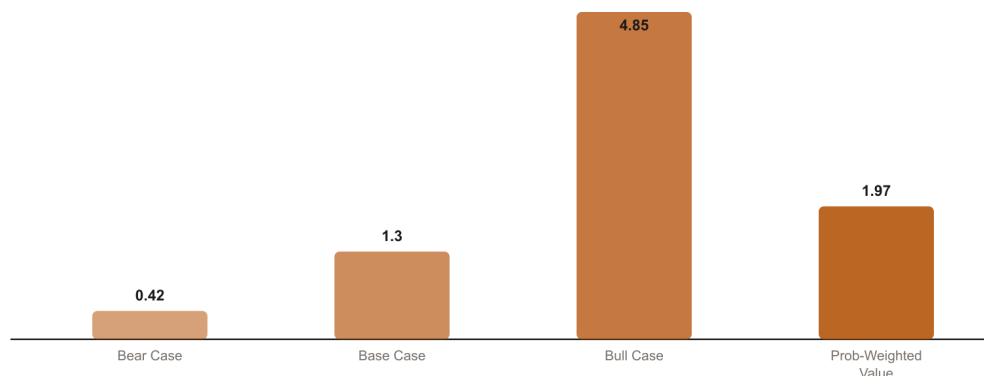
Because Excel data tables recalculate whenever any precedent cell changes, updating your revenue forecast or capital structure assumption instantly refreshes all 49 cells. This makes the table a live scenario dashboard rather than a static snapshot—a critical advantage during live negotiations or management presentations.

Common mistake: Placing the output formula inside the data table range rather than at its top-left corner. Excel will overwrite it during recalculation, destroying your formula. Always put the linking formula at the exact intersection of the row and column headers, outside the body of the grid.

Limit each axis to 6–8 values to keep the printed table on one page and the color gradient meaningful. A 10×10 grid compresses cells to the point where the heat map loses visual clarity and the table spills onto a second page in most report formats.

Break-Even and Goal-Seek Analysis

Scenario Equity Values vs. Weighted



Wide \$0.42–\$4.85 spread signals scenario selection drives conclusion more than WACC precision

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Goal Seek and break-even analysis answer the most practical question in financial modeling: what has to be true for this deal to make sense? Rather than simply accepting a base-case output, you back-solve for the critical input value that sits exactly on the knife's edge between value creation and value destruction. Mastering this technique transforms you from a passive model reader into an active decision-maker.

How Goal Seek Works

Excel's Goal Seek (Data ! What-If Analysis ! Goal Seek) accepts three inputs: the output cell you want to hit (Set Cell), the target value you want it to reach (To Value), and the single input cell Excel is free to change (By Changing Cell). The engine iterates that input using Newton's method until the output converges on your target, typically within milliseconds. The result is a precise break-even figure rather than a range interpolated from a data table.

The canonical financial application is finding the break-even WACC, the discount rate $WACC^*$ at which NPV equals exactly zero:

$$NPV = 0 \Rightarrow \sum_{t=1}^T \frac{CF_t}{(1+WACC^*)^t} - \frac{Investment}{(1+WACC^*)^0} = 0$$

Once you have $WACC^*$, compare it to your actual cost of capital. If $WACC^* = 14\%$ and your firm's WACC is 9%, you have a 500 basis-point margin of safety — the project can absorb significant risk before it destroys value. A thin margin (say, 50 bps) should trigger deeper scrutiny.

The same logic applies to acquisitions. Break-even revenue growth answers "what growth rate must this target sustain to earn its purchase price?" Framing the answer as a premium or discount to the base case — "the deal breaks even at revenue growth 3 percentage points below our base case" — communicates risk intuitively to non-technical stakeholders.

Worked Example: Minimum Exit Multiple in an LBO

Setup. A private equity firm acquires a business for 8× entry EBITDA. Year-0 EBITDA is \$50 million, implying an enterprise value of \$400 million. The capital structure is 60% debt (\$240 million at 7% interest) and 40% equity (\$160 million). Over five years, EBITDA grows at 5% per year, annual free cash flow after interest and taxes is used to repay debt, and the firm targets a 20% IRR on its equity investment.

Simplified cash flows (equity perspective).

Year	EBITDA (\$M)	Debt Remaining (\$M)
0	50.0	240.0
5	63.8	168.0 (after repayment)

Equity at exit = Exit EV " Remaining Debt = (Exit Multiple × \$63.8M) " \$168M

Step 1 — Set up the IRR formula. In Excel, cell B2 holds the exit multiple (start with a guess of 10×). Cell B10 calculates equity proceeds at exit: = B2 * 63.8 - 168. Cell B11 calculates IRR using Excel's IRR function over the cash flow range: Year-0 equity outflow of " \$160M and Year-5 equity inflow from B10.

Step 2 — Launch Goal Seek.

- Go to Data !' What-If Analysis !' Goal Seek.
- Set Cell: B11 (the IRR output).
- To Value: 0.20 (your 20% IRR hurdle).
- By Changing Cell: B2 (the exit multiple).
- Click OK.

Step 3 — Read the result. Excel converges on an exit multiple of approximately 9.7×.

Verification: Equity proceeds = $9.7 \times \$63.8M - \$168M = \$619M - \$168M = \$451M$. IRR check: $(\$451M / \$160M)^{(1/5)} - 1 = (2.819)^{0.2} - 1 \approx 20.0\%$.

Step 4 — Sanity-check against the data table. Pull up your two-variable IRR sensitivity table (entry multiple on one axis, exit multiple on the other). At entry = 8× and exit = 9.7×, the table cell should read 20%. If it reads 19% or 21%, your model has a circular reference or a hardcoded value overriding a formula — fix it before presenting.

Step 5 — Express as a premium to base case. Your base-case exit assumption was 10×. The break-even multiple of 9.7× represents a 3% discount to that assumption. Communicated to the investment committee: "We need only a modest multiple contraction tolerance — the deal breaks even if exit multiples compress by just 0.3 turns."

Extending to Solver

When a problem has multiple unknowns or bounded constraints — for example, finding the minimum revenue growth and minimum margin expansion that jointly deliver a target return — Excel's Solver (Data !' Solver) generalizes Goal Seek. You define an objective cell, list multiple changing cells, and impose bounds (e.g., margin cannot exceed industry maximum of 25%). Solver uses linear programming or GRG Nonlinear methods depending on model structure.

Common mistake: Accepting Goal Seek's answer without checking whether the changing cell has landed on an unrealistic value. A back-solved exit multiple of 22× or a break-even WACC of 1.2% signals a model error, not a genuine insight. Always confirm the result is economically plausible and consistent with your sensitivity table before drawing conclusions.

Variance-Based and Global Sensitivity Methods

One-at-a-time (OAT) sensitivity analysis—changing a single input while holding all others fixed—is intuitive and easy to implement in a spreadsheet. For many simple models it is a perfectly reasonable starting point. However, OAT analysis carries a fundamental blind spot: it cannot detect interaction effects between inputs. In a project DCF, revenue growth and operating margin are rarely independent. A high-growth scenario often compresses margins as the business scales aggressively; a low-growth scenario may preserve margins through cost discipline. When these two inputs move together, their combined effect on NPV is not simply the sum of their individual effects. OAT analysis will systematically misattribute that combined effect, giving the analyst a distorted picture of where risk actually lives.

Why Variance-Based Methods Are More Powerful

Variance-based sensitivity analysis, formalized through Sobol indices, addresses this limitation directly. The core idea is to decompose the total variance of the model output into contributions from each input individually and from every possible combination of inputs:

$$\text{Var}(Y) = \sum_i \alpha_i^2 + \sum_{i < j} \alpha_{ij}^2 + \dots \text{ where } \alpha_i = \text{Var}[E(Y|X_i)]$$

Here, α_i is the first-order Sobol index numerator: it measures how much of the output variance would be eliminated if input X_i were fixed to its true value. The ratio $S_i = \alpha_i / \text{Var}(Y)$ is the first-order Sobol index, ranging from 0 to 1. Higher-order terms α_{ij} capture the additional variance explained only when inputs X_i and X_j vary together—the interaction effect that OAT misses entirely. A model where the sum of first-order indices is close to 1.0 has little interaction; a model where that sum is well below 1.0 is dominated by interactions, and OAT analysis would be dangerously misleading.

Monte Carlo Simulation as the Engine

Computing Sobol indices—and more generally exploring a model's full output distribution—requires Monte Carlo simulation. Instead of evaluating the model at a handful of hand-picked scenarios, Monte Carlo draws thousands of random input combinations simultaneously from specified probability distributions, runs the model for each combination, and assembles the resulting outputs into a distribution.

Choosing the right distribution matters:

- Use a triangular distribution when subject-matter experts can specify a minimum, most-likely, and maximum value but cannot justify a parametric form. It is bounded and transparent.
- Use a lognormal distribution for growth rates and other strictly positive quantities; it prevents the simulation from drawing economically impossible negative growth rates and reflects the multiplicative nature of compounding.
- Use a normal distribution when the input is symmetric and unbounded, such as a discount rate estimated from market data.

Step-by-Step Procedure

- Specify distributions for every uncertain input. Document the mean, standard deviation, and distribution family for each.
- Build the base model in Excel (or your preferred tool) so that every output cell is a pure function of input cells.
- Generate random draws. In Excel, use a Data Table linked to a RAND() cell to force recalculation across thousands of rows, or use a dedicated add-in such as @RISK or Crystal Ball for greater control and reproducibility.

- Record the output (e.g., NPV or enterprise value) for each trial.
- Summarize the output distribution: mean, standard deviation, 5th percentile, 95th percentile, and the probability that the output exceeds a threshold of interest.
- Estimate sensitivity indices by computing the variance of the output conditional on each input (Sobol) or by running rank-correlation analysis as a practical approximation.

Worked Example: Project DCF with 10,000 Trials

Consider a five-year project DCF. Revenue in Year 1 is \$80 M, growing annually at a rate drawn from $N(10\%, 3\%)$ (normal, mean 10%, standard deviation 3%). The WACC is drawn from $N(9\%, 1\%)$. Operating margin is fixed at 25% for simplicity, and terminal value uses a 2% perpetuity growth rate.

Trial-level calculation (one representative draw):

- Drawn growth rate: 11.4%; Drawn WACC: 8.6%
- Year 1–5 revenues: \$80 M, \$89.1 M, \$99.3 M, \$110.6 M, \$123.2 M
- Operating cash flows (25% margin): \$20.0 M, \$22.3 M, \$24.8 M, \$27.7 M, \$30.8 M
- Terminal value at end of Year 5: $\$30.8 \text{ M} \times 1.02 / (0.086 - 0.02) = \475.6 M
- NPV of cash flows + terminal value discounted at 8.6%: \$358.2 M

After 10,000 such trials, the output distribution might look like:

| Statistic | Value |

|---|---|

| Mean NPV | \$312 M |

| Std. Dev. | \$74 M |

| 5th Percentile | \$193 M |

| 95th Percentile | \$448 M |

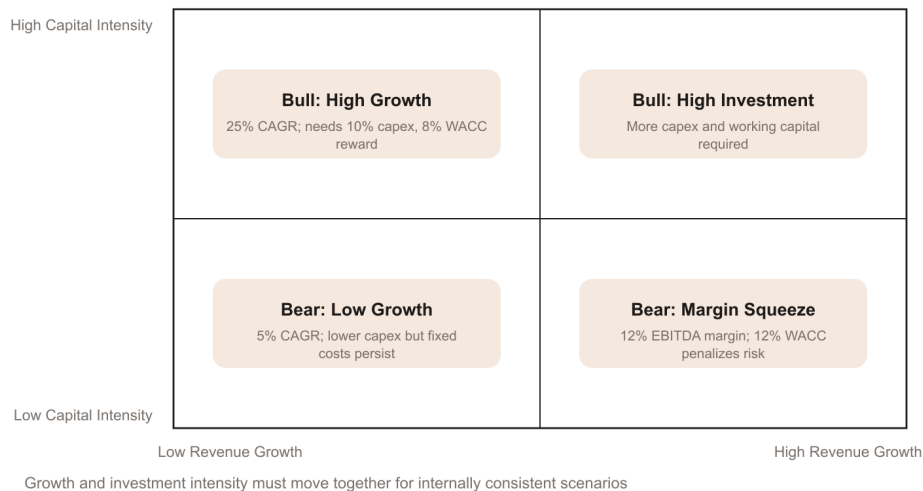
| P(NPV > \$50 M) | 98.7% |

The probability that the project exceeds the \$50 M value threshold is approximately 98.7%—a far richer answer than any single-scenario OAT run could provide.

Common mistake: Analysts often assign normal distributions to growth rates without checking whether the left tail produces negative values. A growth rate of $N(10\%, 3\%)$ will draw values below 2% roughly 0.1% of the time—rare but possible, and potentially nonsensical in context. Use a lognormal or truncated normal distribution when negative values are economically impossible, and always inspect the minimum drawn value across your trials before reporting results.

Building Interactive Scenario Dashboards in Excel

Scenario Assumption Trade-offs



Growth and investment intensity must move together for internally consistent scenarios

Building an interactive scenario dashboard transforms a static DCF model into a living decision-support tool. Instead of maintaining three separate spreadsheets or manually swapping numbers before each meeting, you create one controlled environment where a single dropdown drives every chart, KPI, and output on the page. The six building blocks below work together to make that happen.

Storing Named Input Sets with Scenario Manager

Begin by opening Data !> What-If Analysis !> Scenario Manager and creating three named scenarios: Base, Bull, and Bear. For each scenario, designate the same set of changing cells — revenue growth rate, EBITDA margin, terminal growth rate, and WACC — and enter the appropriate values. Once all three are saved, click Summary to generate a comparison table on a new sheet. This table is a permanent audit trail showing every assumption side by side, which is invaluable during presentations when stakeholders ask "what changed?"

Pulling Active Assumptions with Named Ranges and INDEX/MATCH

Create a structured AssumptionsTable on a hidden Inputs sheet. Rows represent individual inputs (revenue growth, margin, WACC, etc.) and columns represent scenarios. Name the range AssumptionsTable, name the column headers range Scenario_Names, and assign each input row a number stored in a constants block.

The core retrieval formula is:

,

ActiveAssumption = INDEX(AssumptionsTable, MATCH(DropdownSelection, ScenarioNames, 0), Input_Row)

,

Every live model cell references Active_Assumption cells rather than hard-coded numbers, so the entire model recalculates the moment the dropdown changes.

Dropdown List via Data Validation

On the dashboard sheet, select the scenario-selector cell (e.g., B2) and apply Data !' Data Validation !' List, pointing the source to ScenarioNames. Label it clearly: "Select Scenario:". This single cell becomes DropdownSelection in every INDEX/MATCH formula throughout the workbook.

Common mistake: Analysts sometimes type scenario names directly into the validation source box (e.g., Base,Bull,Bear) instead of referencing the named range. If you later rename a scenario in the Assumptions_Table header, the dropdown won't update automatically — always point to the range.

Fully Worked Numerical Example

Setup: A consumer goods company with a five-year DCF. The Assumptions_Table holds the following values:

Input	Bear	Base	Bull
Revenue Growth (%)	3.0	6.0	10.0
EBITDA Margin (%)	14.0	18.0	22.0
WACC (%)	10.5	9.0	8.0
Terminal Growth (%)	1.5	2.5	3.0

Step 1 — User selects "Base" from the dropdown in cell B2.

Step 2 — Retrieve Revenue Growth. Input_Row for revenue growth = 1.

=INDEX(AssumptionsTable, MATCH(B2, ScenarioNames, 0), 1)

MATCH finds "Base" in column 2 of Scenario_Names, so the formula returns 6.0%.

Step 3 — Build Year 1–5 revenues. Base year revenue = \$500 M. Year 1 = $500 \times 1.06 = \$530$ M; Year 2 = $530 \times 1.06 = \$561.8$ M, and so on through Year 5 = \$669.1 M.

Step 4 — Calculate EBITDA. EBITDA Margin retrieved = 18.0%. Year 1 EBITDA = $530 \times 0.18 = \$95.4$ M.

Step 5 — Discount free cash flows at WACC = 9.0% and add terminal value. Assume FCF conversion at 55% of EBITDA. Year 1 FCF = $95.4 \times 0.55 = \$52.5$ M; PV = $52.5 / 1.09 = \$48.2$ M. Repeat for Years 2–5, then add PV of terminal value. Total Enterprise Value = \$820 M.

Step 6 — Compute KPIs. Net debt = \$120 M; shares outstanding = 50 M.

- Equity Value = $820 - 120 = \$700$ M !' \$14.00 per share
- EV/EBITDA = $820 / (669.1 \times 0.18) = 820 / 120.4 = 6.8\times$

These two figures populate the KPI summary box automatically because those cells reference the live model outputs.

Dynamic Combo Chart

Insert a combo chart referencing the live revenue and EBITDA margin rows. Set revenue as a clustered column series (primary axis) and EBITDA margin as a line (secondary axis). Because both series point to calculated output cells — not the Assumptions_Table directly — switching the dropdown from Base to Bull instantly redraws the chart with Year 1–5 revenues climbing to \$805 M and margins expanding to 22%.

KPI Summary Box Layout

Arrange four metrics in a 2×2 grid: equity value per share, implied EV/EBITDA, IRR, and payback period. Each cell contains a simple reference to the corresponding model output. For a side-by-side comparison, add a static column pulling from the Scenario Manager Summary table so stakeholders can see all three scenarios simultaneously while the dynamic column reflects the dropdown selection.

Protecting the Dashboard

Select all formula and output cells, open Format Cells !' Protection, and check Locked. Leave only the dropdown cell and any permitted input cells unlocked. Then apply Review !' Protect Sheet with a password. During a live presentation, accidental keystrokes cannot overwrite a formula, and the model retains its integrity throughout the meeting.

| Glossary

- Scenario Analysis — Evaluating model outputs under multiple internally consistent sets of assumptions simultaneously.
- Sensitivity Analysis — Measuring how much a model output changes when one or more inputs vary.
- Tornado Chart — Horizontal bar chart ranking inputs by their impact range on a single output.
- Data Table — Excel feature computing an output across a grid of one or two changing input values.
- Monte Carlo Simulation — Repeated random sampling of input distributions to generate a probabilistic output distribution.
- Sobol Index — Variance-based measure of each input's fractional contribution to total output variance.
- Goal Seek — Excel tool that back-solves one input cell to force a formula cell to a target value.
- Terminal Growth Rate — Perpetual growth rate applied to final-year free cash flow in a DCF terminal value calculation.

